

NGUYEN CAO XUAN PHUC

INNOVATION & TECHNOLOGY ENTHUSIAST | STUDENT RESEARCHER | SOCIAL
IMPACT LEADER

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PROFILE

I am a high school student interested in innovation, technology, and sustainable development. Through interdisciplinary projects in AI, environmental protection, assistive technology, and digital research, I have developed experience in teamwork, project coordination, communication, and problem-solving. I hope to combine technology with business and social responsibility to create practical solutions that contribute to communities and long-term sustainable development.

EDUCATION & ACADEMIC CREDENTIALS

Nguyen Hue High School for the Gifted

Hanoi, Vietnam

Specialized Physics Program | Grade 11

Academic record: Grade 10 GPA 9.1/10; Grade 11 GPA 9.2/10

SAT 1440(math 800/800) | IELTS 7.5

AP Physics C: 5/5

SELECTED HONORS & AWARDS

Gold Award - World Invention Creativity Olympic (WICO 2026)

Jul 2026 | Seoul, Korea

Project: SafeStride - A High-Precision Spatial Awareness and Priority Alert System for the Blind on Smartphones

- Developed an assistive smartphone concept designed to improve spatial awareness and deliver priority alerts for blind users.
- Awarded Gold; the certificate recognizes the team's presentation quality, creativity, and innovation.

First Prize, Arts & Technology Award, and Global Nominee - NASA

Oct 2025 | Ninh Binh

International Space Apps Challenge

Team BlueMarble | Project: BloomWatch

- Created an Earth-observation learning platform that transforms NASA satellite data on global flowering phenomena into an accessible multimedia experience.
- Integrated data research, AI-generated characters, visual storytelling, and collaborative product development during a 48-hour hackathon.

Third Prize - Environmental Protection Ideas in the Transport Sector
Competition

Jun 2026 | Hanoi

University of Transport and Communications | Project: ECOMeSort

- Designed an AI-enabled mobile application that guides waste sorting, connects users within the recycling ecosystem, and promotes environmental action through green-community features.

Build Award - Vietnam VEX V5 Robotics National Championship

2026 | National Round

Member, NGS Hogrider

- Contributed to a team recognized for robot construction quality, reliability, craftsmanship, and competition readiness.

International Academic Awards

2021-2022

- Bronze Award - Thailand International Mathematical Olympiad (TIMO), Heat Round 2021-2022.
- Bronze Award - Philippine International Mathematical Olympiad (PIMSO), Heat Round 2022.
- Merit Award, National Level - ASMO English 2022.

RESEARCH EXPERIENCE & SCHOLARLY ENGAGEMENT

Research Intern (Ongoing) - Institute of Information Technology, VAST

18 May-24 Sep 2026 |
Hanoi

AI Systems Department, Vietnam Academy of Science and Technology

- Participating in a structured research internship focused on applied AI systems, while gaining experience with scientific workflows, technical documentation, and interdisciplinary collaboration.

Student Author and Presenter - VNICT 2025, 28th National Conference on Selected ICT Problems

Nov 2025 | Ninh Binh

- Co-authored and presented a conference paper on an improved dual-image reversible data-hiding method designed to increase embedding capacity for digital-content authentication.
- Developed early experience in technical writing, experimental reasoning, academic presentation, and peer-facing scientific communication.

Selt study :Completed and received certificates for the following Coursera courses:

- AI Fundamentals
- Vibe Coding with Claude Code
- AI for Data Analysis
- Vibe Coding with GitHub Copilot

LEADERSHIP, ENTREPRENEURSHIP & SOCIAL IMPACT

Founder & Project Chair - ECOMe 2025-Present

Student-led environmental education and community-action initiative

- Founded and led awareness campaigns reaching approximately 1,800 people through field activities in the Mekong Delta and multiple Vietnamese provinces, including Hai Duong and Quang Ninh.
- Led a June 2026 campaign that combined environmental communication with community service at six orphanages and child-care centers, supporting projects such as clean-water wells, rural bridges, street lighting, community kitchens, and livelihood assistance.
- Initiated and chaired an environmental-ideas competition; coordinated communications, participant engagement, and project promotion.
- Project evidence: [ECOMe activity page](#)

Community Service Volunteer and Campaign Organizer 2025-2026

- Participated in charitable programs across Bac Ninh, Hai Duong, Lang Son, Dak Lak, the Mekong Delta, and other localities.
- Supported two 2025 campaigns that delivered a combined VND 115 million in gifts and aid; also joined relief efforts serving 225 households affected by natural disasters in Dak Lak.

STEM PROGRAMS, INTERNATIONAL EXPOSURE & SCHOOL LEADERSHIP

Open Data Exploration 2026 Summer Camp - Organizing Team Member Jul 2026 | American Center, U.S. Embassy Hanoi

- Served in two functions: Communication & Design and Training & Data Research, supporting program delivery, educational content, and participant engagement.

Robotics Summer Camp 2026 - Mechanical Engineering Team Member Jul 2026 | American Center, U.S. Embassy Hanoi

- Supported free robotics education for Vietnamese students through technical teamwork and hands-on camp activities.

AI for Good Vietnam 2026 - Team ECOMeSort May 2026 | Hanoi

- Participated in an education program focused on applying AI to socially beneficial challenges.

Campus Tour and Learning Activities - University of Tsukuba Jul 2025 | Japan

- Completed academic and cultural learning activities hosted by the university's international student support office.

Media & Community Roles 2024-2026

- Media Member, CNH Basketball Club - Panthers (2024-2025 and 2025-2026).
- Media Department Collaborator, Net Dep Khoi Chuyen 2025.
- Volunteer, Japanese Festival in Hanoi 2025.

CORE COMPETENCIES & ACADEMIC INTERESTS

Research & Innovation

Scientific writing; reversible data hiding; digital-content authentication; problem formulation; evidence-based presentation.

Leadership & Communication

Project management; team collaboration; public outreach; media strategy; visual design; event organization.

Technology & Product Thinking

AI-enabled applications; assistive technology; robotics; data-driven storytelling; human-centered solution design.

Languages

Vietnamese (native); English (IELTS Academic 7.5).